Anjana Mohandasan

Personal Data

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Publications:	ADS Library

Academic Timeline

Aug 2024 - now

Postdoctoral researcher in Astrophysics
Study of ultracool dwarf stars with *Euclid* and *Gaia* space telescopes.
Supervisor: Prof. Céline Reylé
Besançon Observatory, UTINAM Institute, OSU THETA, Université Marie et Louis Pasteur
41 bis avenue de l'Observatoire, BP 1615, 25010, Besançon, France

Oct 2020 - Apr 2024

Doctor of Philosophy (PhD) in Astronomy
Department of Physics and Astronomy, University of Padova, Padova
Thesis - 'Photometric binaries in Magellanic Cloud star clusters'
Advisor: Prof. Antonino P. Milone
Defended on 05 Apr 2024
- Department Scholarship for Doctoral studies

July 2018 - Oct 2020

Master of Science in Physics
School of Physics, University of Hyderabad
9.43/10 | First Class with Distinction | Grade: A+
Thesis - 'Quantum Mechanics in Non-inertial Frame'
Advisor: Prof. E. Harikumar
- Prof. M. Shakuntala Memorial Medal.

Jun 2015 - Mar 2018

Bachelor of Science in Physics
St. Joseph's College, affiliated to Calicut University
95.6/100 | First Class With Distinction | Grade: A+
Thesis - 'Revisiting Newton's Law of Gravitation'

Jun 2013 - Mar 2015

Higher Secondary Education in Physics, Maths, Chemistry, and Biology.
Providence Girls Higher Secondary School
99.33/100 | First Class With Distinction | Grade: A+
- Inspire Scholarship for Higher Education in Natural Sciences.
- Pratibha Scholarship for Higher Education in Natural Sciences (declined).

Publication List

First authored

(i) 'Euclid: Quick Data Release (Q1) – Ultracool dwarfs in the Euclid Deep Fields'
Euclid Collaboration: **A. Mohandasan**, R. L. Smart, C. Reyle, et al.
A&A 713, A245 (2026) | DOI: 10.1051/0004-6361/202660157
→ *Lead analysis on early mission space spectroscopy from the Euclid Space Telescope.*

(ii) 'Photometric Binaries in 14 Magellanic Cloud star clusters'
Mohandasan, A., A. P. Milone, G. Cordoni, et al.
A&A 681, A42 (2024) | DOI: 10.1051/0004-6361/202347424
→ *High-precision binary star characterisation using Hubble Space Telescope photometry.*

(iii) 'Photometric Binaries in Magellanic Cloud star clusters'
Anjana Mohandasan | Zenodo Open Research Data | DOI: 10.5281/zenodo.10939691

(v) 'Exploring ultracool dwarfs with Euclid and Gaia'
Anjana Mohandasan, R. L. Smart, C. Reylé
submitted to **Astronomical Notes (Astronomische Nachrichten)**

(v) 'Anomalous population in Omega Centauri: Binary origin'
Mohandasan, A., E. Pancino, L. Monaco, et al.
(In preparation, advanced draft for submission to **A&A**)

Co-authored

(vi) 'Gaia Data Release 4: The Bottom of the Main Sequence'
Gaia Collaboration, L. M. Sarro, R. L. Smart, **A. Mohandasan**, et al.
ESA Flagship sSpace mission Performance Verification Paper, submitted to **A&A**

(vii) 'Gaia Data Release 4: Validation and Technical Report'
Gaia Collaboration, et al. (incl. **A. Mohandasan**)
Consortium Performance Validation Paper, submitted to **A&A**

Additional Co-authored Works: Complete bibliographic record available via the NASA ADS Library: **'Research_Papers_Anjana_Mohandasan'**

Observational Experience

GTC Observation — Conducted three nights of GTC/EMIR observation from the 7th to the 10th of June 2025. Co-I of an approved Gran Telescopio Canarias (GTC) observational Proposal for 6 nights in EMIR and OSIRIS modes.

JWST Observation Time — Co-I of an approved JWST observational proposal of observation time 21 hrs (Proposal ID: 9012 | Cycle: 4 | Year: 2025).

HST Observation Time — Co-I of an approved HST observational Proposal (Proposal ID: 17075 | Cycle: 30 | Year: 2022).

Conference Talks

13 Oct 2026 – 15 Oct 2026 — Invited seminar - 'Exploring Ultracool dwarfs with Euclid'
ARIES, Nainital, India

10 Aug 2026 – 14 Aug 2026 — Contributed talk - 'Exploring Ultracool dwarfs with Euclid'
The conference Substellar Astrophysics
Tordesillas, Spain.

28 Jun 2026 – 03 Jul 2026 — Contributed talk - 'Exploring Ultracool dwarfs with Euclid'
The conference The annual meeting of the European Astronomical Society (EAS) of the year 2026
SwissTech Convention Centre, Lausanne, Switzerland.

25 May 2026 – 29 May 2026 — Contributed talk - 'Ultracool dwarfs in the era of Euclid' in EC-UCD splinter.
The annual meeting of the Euclid Consortium of the year 2026, Barcelona, Spain.

01 Dec 2025 – 03 Dec 2025 — Invited talk - 'Exploring Ultracool dwarfs with Euclid'
The conference Working with Euclid Q1: From Data Access to Science.
European Space Astronomy Centre (ESAC), Madrid, Spain.

17 Nov 2025 – 19 Nov 2025 — Contributed talk - 'Exploring Ultracool dwarfs with Euclid'
The conference France for Euclid 2025: Symposium Euclid France 2025.
University of Strasbourg, Strasbourg, France

01 Sep 2025 – 05 Sep 2025 — Contributed talk - 'Exploring Ultracool dwarfs with Euclid'
The conference BD30: Brown dwarfs keep their cool: Thirty years of substellar science.

	La Gomera, Canary Islands, Spain.
23 Jun 2025 – 27 Jun 2025	Contributed talk - 'Ultracool dwarfs in the era of Euclid' The annual meeting of the European Astronomical Society (EAS) of the year 2025. University College of Cork, Cork, Ireland.
11 Jun 2025 – 18 Jun 2025	Seminar - 'Ultracool dwarf study with Euclid' IAC, Tenerife, Spain.
24 Mar 2025 – 28 Mar 2025	Contributed talk - 'Ultracool dwarfs in the era of Euclid' in EC-UCD splinter. Contributed talk - 'Photometry of point sources' in EC-Photometry splinter. Poster - 'Ultracool dwarfs in the era of Euclid' in EC symposium. The annual meeting of the ESLAB and Euclid Consortium of the year 2025, Leiden, The Netherlands.
7 Jan 2025 – 9 Jan 2025	Contributed talk - 'Ultracool dwarfs in the era of Euclid' The annual meeting of the Euclid - France of the year 2024 CEA Saclay, Paris, France.
7 Dec 2024 – 9 Dec 2024	Contributed talk - 'Photometric binaries in globular clusters' Astrocoffee, Observatoire de Strasbourg, Strasbourg, France.
7 Oct 2024 – 11 Oct 2024	Contributed talk - 'Binaries and related stellar populations in star clusters: An exploration with photometry' The conference Resolved stellar populations from photographic plates to large surveys: A celebration of Peter B. Stetson's career in astronomy FMF Centre, Florence, Italy
10 Jul 2023 – 14 Jul 2023	Contributed talk - 'Photometric Binaries in Extragalactic Star Clusters' The annual meeting of the European Astronomical Society (EAS) of the year 2023 ICE congress centre, Kraków, Poland.
3 Jul 2023 – 7 Jul 2023	Contributed talk - 'Photometric Binaries in Magellanic Cloud Globular Clusters' The conference - 'A Multiwavelength View on Globular Clusters Near and Far: From JWST to the ELT' Bolzano, Italy.
14 Nov 2022 – 18 Nov 2022	Contributed talk - 'Photometric Binaries in Extragalactic Globular Clusters' The conference - 'The Impact of Binaries on Stellar Evolution' Max Planck Institute for Astrophysics, Garching, Germany.
12 Sep 2022 – 16 Sep 2022	Contributed talk - 'Photometric Binaries in Galactic and Extragalactic Globular Clusters' The conference - 'Wheel of Star Formation' Charles University, Prague.

Selected Grants, Distinctions and Awards

Research Grants	• **CNES APR Research Funding (2026):** Competitive research grant awarded by CNES (Centre National d'Études Spatiales) to support scientific activities and *Euclid* mission collaborations.
Scholarships	• Department Scholarship for Doctoral Studies, University of Padova, Italy, 2020. • Junior Research Fellowship, Government of India, 2019 (declined). Awarded via CSIR UGC NET Physics Exam; eligible for fully funded doctoral studies in any Indian institute. • PhD fellowship, IISER, Pune - 2020 (declined). • PhD fellowship, IIST, Trivandrum - 2020 (declined). • Summer Research Fellowship, Indian Academy of Sciences, 2019. • Inspire Scholarship, Government of India, 2015. Awarded to the top 1% performers in the Indian Higher Secondary School Examination.

• Prathibha Scholarship, 2015. Awarded to the top 100 students out of 0.2 million in the science stream, Kerala State Higher Secondary School Examination (declined).
• Other scholarships: District Merit Scholarship (2013), Inculcate Scholarship (2011), Upper School Scholarship (2010), Lower School Scholarship (2007).

Awards

• Prof. M Shakuntala Memorial Medal, University of Hyderabad, 2020. For academic excellence in M.Sc. Physics.
• First rank, Pondicherry University M.Sc. Physics Entrance Exam, 2018.
• A grade, awarded for Maths group project, Kerala State Maths Fair, 2014.
• Maths Talent Search Exams. State ranks: 6th (2012), 8th (2011).
• Physics Talent Search, State rank: 17 (2015, obtained in the first year of graduate studies).
• 'Galileo Little Scientist' State-level Quiz Competitions, 2010.

Consortium and Memberships

Euclid Consortium

Member of the Milky Way resolved stellar population science working group and SPE pipeline OUs of DSS within the *Euclid* consortium. I work on the Ultracool dwarfs work package. Member of the **Early Career Committee** of *Euclid* Consortium, coordinating the Mentoring programme.

Gaia DPAC

Member of the *Gaia* Data Processing and Analysis Consortium. Involved in providing validation of *Gaia* catalogues, as part of CU9. Responsible for building the LF of the project on DPAC211, about stars at the bottom of the main sequence (Sarro et al. 2026).

IAU

International Astronomical Union's junior member since July 2026. Participation in the early career researchers working group.

LSST / *WEAVE*

External or consortium-associated participant in LSST and WEAVE survey science discussions.

Mentoring, Outreach and Community Leadership

Feb 2026 – present

'English Hour'
Initiated and lead weekly sessions designed to improve oral communication and discussion skills among an international community of academic researchers and administrative staff, fostering an inclusive environment for cross-cultural dialogue and workplace well-being.

Oct 2026 – present

Euclid Consortium Early Career Committee
Member of the Early Career Committee, co-coordinating the Mentoring Programme for early-career researchers across the international *Euclid* Consortium.

Oct 2024 – Jun 2025

Master's Project Supervision
Primary co-mentor for Mr. Muhammed Nabyan Arken's Master's research project, UTINAM / UMLP, Besançon.

2024 – present

Master's Thesis Referee
Appointed referee and reviewer for Master's thesis defences (Mr. Tituan Alphonse and Mr. Sohan Sengupta), UMLP, Besançon.

21 Oct 2023

'Mimimi Mico Mama Macro' – Art & Astronomy Outreach
Co-organized a public art-meets-astronomy initiative across Venice, Italy. Published an outreach magazine and delivered a performative public lecture on stellar astrophysics. Supported by the Swiss Arts Council, Pro Helvetia.

Programming and Data analysis Skills

Programming Languages:	Python | FORTRAN | SuperMongo | LaTeX | ADQL
Data Analysis Tools:	ESA Datalabs and Archives | MAST | CDS services | SPLAT | iSpec | TOPCAT | DS9 | IRAF | Origin | Aladin |GravPot| various Python and Linux-based tools
Astronomical Techniques:	Photometry | Spectroscopy | Model Fitting | Radial Velocity Analysis | Catalogue Cross-Matching
Instruments and Surveys:	Data reduction and/or analysis of observations from HST | Euclid | Gaia | GTC | MUSE | SPHEREx; extensive experience handling catalogues from multiple ground and space-based instruments.

Other training and schools

07 Sep 2026 – 11 Sep 2026	Gaia DPAC meeting 2026 University of Geneva, Switzerland.
02 Mar 2026 – 06 Mar 2026	Gaia MW Educado training school: Astro AI and Machine Learning University of Ghent, Belgium.
21 Sep 2025 – 26 Sep 2025	Evry Schatzman School 2025: Stellar Physics in the PLATO era The school, sponsored by PNPS, CNRS, CNES, and others, was held at the Roscoff Marine Station in Aussois, France.
30 Sep 2024 – 4 Oct 2024	Evry Schatzman School 2024: Disks and planetary formation in the JWST and ELT era The school, sponsored by PNPS, CNRS, CNES, and others, was held at the Roscoff Marine Station in Brittany, France.
28 Aug 2022 – 3 Sep 2022	International Summer School, Life and Death: From Stars to Compact Objects Conducted by the University of Padova and took place in the University Observatory in Asiago, Italy.
17 Oct 2022 – 21 Oct 2022	Astro Hack Week 2022 Conducted at the Max Planck Institute for Astronomy, Heidelberg.
6 Jun 2022 – 17 Jun 2022	Summer School in Astrostatistics and Astroinformatics Conducted by the Centre for Astrostatistics at the Pennsylvania State University (held online).
12 Jul 2021 – 23 Jul 2021	International Summer School (held online) - 'The Interstellar Medium of Galaxies, from Epoch of Reionisation to the Milky Way'.
21 Jun 2021 – 26 Jun 2021	Rencontres Exobiologiques Pour Doctorants - 2021 (held online)
May 2019 – Jun 2019	Summer Research Intern Project - 'Photometric Binaries in Galactic Open Clusters' Advisor - Prof. Annapurni Subramaniam Indian Institute of Astrophysics, Bengaluru, India - Summer Research Fellowship by the Indian Academy of Sciences.
Jun 2017	Summer Research Intern Project - 'Study of X-ray Detectors using Radioactive Sources' Advisors: Dr. Tapan Nandi and Mr. Deepak Swami Inter-University Accelerator Centre, Delhi, India.
Jun 2014 - Dec 2014	Project Student in Mathematics Project - 'Navigation through Composition' - The project was awarded an A grade in the Kerala State Maths Fair - 2014.

Language Proficiency

Malayalam:	Mother tongue
English:	Fluent
Hindi:	Advanced
Italian, French:	Basic

References

Prof. Céline Reylé — Astronomer (celine@obs-besancon.fr)
Besançon Observatoire, OSU THETA, UTINAM Institut, Université Marie et Louis Pasteur, 41 bis avenue de l'Observatoire, BP 1615, 25010, Besançon, France
Relation: Postdoc supervisor

Prof. Antonino P. Milone — Associate Professor (antonino.milone@unipd.it)
University of Padova Vicolo dell'Osservatorio, 3, 35122, Padova, Italy
Relation: PhD advisor

Dr. Richard L. Smart — Technological Researcher III level (richard.smart@inaf.it)
Osservatorio Astrofisico di Torino, Via Osservatorio, 20, 10025, Pino Torinese TO, Italy
Relation: Research collaborator

Declaration

The information provided here is true to the best of my knowledge.

Place: Besançon, France
Date: 11/10/2026